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EDUCATION

Northeastern University, Boston, MA

Sep 2025 – Apr 2027

Master of Science in Robotics; GPA: 3.98

Relevant Coursework: Robot Mechanics and Control; Mobile Robotics; Robot Sensing and Navigation; Reinforcement Learning

KL University, India

Aug 2021 – May 2025

Bachelor of Technology in Artificial Intelligence and Data Science

Relevant Coursework: Machine Learning; Deep Learning; Natural Language Processing; High Performance Computing; Signal Processing; Computer Vision; Image Processing

RESEARCH & EXPERIENCE

PGTP-Net 3D Extension: Pose-Guided 3D Trajectory Prediction for Mobile Robots

- Extending PGTP-Net to lift pose (full SMPL pose + shape) and trajectory output from 2D to 3D world coordinates, enabling anticipation of behaviors like stair ascent, curb stepping, and vertical motion that 2D models structurally cannot represent.
- Replacing the perception front-end with an end-to-end SMPL regressor (HMR 2.0 / WHAM) and designing a cross-dataset generalization study (JTA → JRDB-Pose3D, Human3.6M → 3DPW) with closed-loop MPC evaluation. Target venues: ICRA / RA-L.

GPS-Denied Drone Navigation on Qualcomm Dragonwing

- Extending the GPS-denied UAV stack to a Qualcomm Dragonwing IQ-9075 EVK (100 TOPS NPU, 36 GB LPDDR5) with PX4/Pixhawk and onboard edge-AI inference.
- Designing a comparative study of four navigation approaches (Visual SLAM, LiDAR-only with FAST-LIO2, LiDAR + camera fusion with FAST-LIVO2, and a Vision-Language-Action baseline), evaluated on ATE/RPE and exploration coverage. Target venues: ICRA / IROS / RA-L.

Multi-View ACT UR5 Servoing: Cable Insertion

Intrinsic AI for Industry Challenge 2026 (Qualification Phase)

- Built a three-tier hybrid manipulation pipeline (ACT imitation policy + multi-view perception + closed-loop visual servoing) for autonomous fiber optic cable insertion (~0.5–1.0 mm clearance) on a UR5e arm, trained ACT on 302 teleoperated episodes and layered geometric perception to close the out of distribution generalization gap.
- Engineered a YOLOv8s-OBB + PnP 6-DoF pose with synthetic auto labeling (5,160 labels in under an hour, mAP50 ≈ 0.965), plus multi view scene registration and a hybrid YOLO/registered pose visual servo with failure recovery.

PROJECTS

GPS-Denied Autonomous Drone Navigation in Confined Environments

Jan 2026 – Apr 2026

- Developed a fully autonomous UAV stack for GPS-denied multi-branch tunnel networks, integrating LIO-SAM LiDAR-inertial SLAM as primary localization with RTAB-Map fallback for featureless corridors, OctoMap probabilistic 3D occupancy mapping, and Nav2 driven frontier based exploration.
- Developed an NIR spectral mineral classification pipeline using kNN/SVM trained on the USGS Spectral Library, Experimented on the DBSCAN clustering on PCD to capture mineral zones in 3D space via SLAM-georeferenced pose estimates

Curiosity-Driven Exploration for Sparse Rewards: Reinforcement Learning Jan 2026 – Apr 2026

- Implemented PPO+RND, DQN+ICM, Rainbow DQN to tackle exploration in one of Atari's most notoriously unsolved sparse reward game (Adventure 2600).
- Engineered RAM-based state extraction, a unified dense reward-shaping framework, and a curriculum learning pipeline, diagnosed and patched recurring failure modes including greedy policy collapse, reward-farming exploits, and incorrect ALE RAM addresses

PGTP-Net (v1): Pose-Guided Pedestrian Trajectory Prediction Sep 2025 – Dec 2025

- Built a dual-stream architecture (Social-LSTM with attention pooling for trajectories + Transformer pose encoder) fused into a CVAE decoder that generates $K = 20$ candidate 4.8-second futures conditioned on 2D/3D pose key points.
- Reduced trajectory error (ADE/FDE) by 35–46% on JTA with 3D ground truth poses and 25–28% on ETH/UCY with estimated 2D poses, validating that skeletal pose carries anticipatory signal point-mass models discard.

6-DOF UR3 Manipulator: Kinematics and Task-Space Trajectory Tracking Sep 2025 – Dec 2025

- Built a forward kinematics model of a 6-DOF UR3 arm from published link geometry, computing joint and end effector positions through geometric and trigonometric relations, with a finite difference numerical Jacobian for joint to Cartesian velocity mapping.
- Developed a damped least squares inverse kinematics solver with adaptive damping for robustness near singularities, tracking hexagonal, elliptical, and lemniscate task space trajectories and visualizing the reachable workspace via a 4,000-sample Monte Carlo simulation in Python/NumPy.

Hybrid Parallel Robot Arm: Parallel Shoulder and Four-Bar Gripper Design Sep 2025 – Dec 2025

- Designed a hybrid parallel robot arm in SolidWorks: a 3-DOF spatial parallel shoulder using spherical-prismatic-spherical (S-P-S) limbs on a hexagonal base for pitch and roll tilt, paired with a four-bar parallel-jaw gripper (≈ 80 mm opening) actuated by Dynamixel RX-64 servos.
- Performed forward-kinematics and mobility analysis relating prismatic limb-length changes to platform orientation, showing that ± 10 mm limb adjustment yields up to $\pm 25^\circ$ platform tilt, with tilt and gripper motion demonstrated in SolidWorks Motion Study.

TECHNICAL SKILLS

Programming: Python (PyTorch, TensorFlow), C++, C, CUDA, ROS, ROS2, Linux, Git, Docker

Hardware: Qualcomm Dragonwing IQ-9075, PX4/Pixhawk, Intel RealSense D435i, UR5e arm, IMU

SLAM & Localization: LIO-SAM, FAST-LIO2, FAST-LIVO2, RTAB-Map, ORB-SLAM3, OctoMap, Kalman filtering, EKF

Planning & Controls: Nav2, frontier-based exploration, EGO-Planner, GBPlanner2, Model Predictive Control, inverse kinematics (DLS)

Perception & Manipulation: 6-DoF pose estimation (PnP, SMPL/HMR/WHAM), YOLO / YOLO-OBB, visual servoing, multi-view scene registration, camera calibration, depth perception

AI & Deep Learning: Reinforcement learning (PPO, RND, Monte Carlo), Action Chunking Transformers (LeRobot ACT), Transformers, CVAE, LSTM, ResNet

Design & Simulation: SolidWorks, Fusion 360, Onshape, Gazebo, RViz